

Experiment 10

Subnetting and Creating Subnet Masks

Lab Objective:

To perform **subnetting** on a given IP address and **create subnet masks** to divide a network into smaller, efficient sub-networks.

What is Subnetting?

- **Subnetting** is the process of dividing a large IP network into **smaller sub-networks (subnets)**.
- It helps in better **IP address management, network performance, and security**.
- Each subnet acts like a separate network but still remains part of the original large network.

Why Subnetting is Important

- Efficient utilization of IP addresses
- Reduces broadcast traffic
- Improves network performance and manageability
- Enhances security by isolating network sections
- Required in large organizations with multiple departments

Understanding IP Classes and Default Subnet Masks

Class	Starting IP	Range	Default Subnet Mask
A	1.0.0.0	1-126	255.0.0.0 (/8)
B	128.0.0.0	128-191	255.255.0.0 (/16)
C	192.0.0.0	192-223	255.255.255.0 (/24)

Performing Subnetting (IPv4)

Suppose you are given:

- IP Address: 192.168.1.0
- Requirement: 4 subnets

Steps:

1. Determine the subnet bits:

- For 4 subnets, you need 2 bits ($2^2 = 4$)

2. Modify subnet mask:

- Default mask: /24 → add 2 bits → New mask: /26
- Subnet Mask: 255.255.255.192

3. Calculate subnet ranges:

Subnet	Network Address	First Host	Last Host	Broadcast Address

1	192.168.1.0/26	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193	192.168.1.254	192.168.1.255

Creating Subnet Masks (Binary to Decimal)

- /26 in binary:
11111111.11111111.11111111.11000000
- In decimal:
255.255.255.192

Subnetting in IPv6

- IPv6 already uses **prefix length** to divide networks (no subnet masks like IPv4)
- Example:
 - IPv6: 2001:db8:abcd::/48 → Want 4 subnets
 - Add 2 bits to /48 → New prefix: /50
 - Subnets:
 - 2001:db8:abcd:0::/50
 - 2001:db8:abcd:4000::/50

- 2001:db8:abcd:8000::/50
- 2001:db8:abcd:c000::/50

Verification Commands

On Windows (IPv4):

ipconfig

ping <other subnet host>

On Cisco Router:

show ip interface brief

To assign IP:

interface GigabitEthernet0/0

ip address 192.168.1.1 255.255.255.192

no shutdown

Outcome:

After completing this experiment, students will be able to:

- Understand and perform subnetting
- Create custom subnet masks
- Divide a network logically using IPv4 and IPv6
- Improve network management and address allocation